FRANCESCO GIACOMARRA | Curriculum Vitae

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• LinkedIn

Summary and background

I am a postdoctoral researcher in trustworthy Generative modeling within the RobustifAI project at the TU Wien. I am particularly interested in developing principled methods to obtain guarantees for deep generative models, and in the exploration of techniques that integrate formal methods approaches to establish if and when a model is compliant with respect to a desired behavior.
I am also active in science communication and enjoy working collaboratively in multidisciplinary research environments.

Current position

TU WIEN
October 2026 - ongoing
PostDoc within the TrustCPS group, lead by Professor Ezio Bartocci Vienna, Austria
Research on Trustworthy Generative AI funded by the RobustifAI project

Education

UNIVERSITY OF TRIESTE
December 2022 - February 2026
PhD in Applied Data Science and Artificial Intelligence (Supervisors: L. Bortolussi and F. Cairoli) Trieste, Italy
Final Grade: PhD awarded cum Laude (with honors)
Main research activities:

- Integration of statistical and deep models for graph-structured data
- Formal guarantees for deep generative models
- Neurosymbolic approaches for reliable scenario generation

UNIVERSITY OF PADUA
October 2020 - September 2022
M.Sc. in Data Science (Curriculum: Mathematics for Data Science) Padua, Italy
Final Grade: 110/110, cum Laude (with honors)

UNIVERSITY OF PADUA
October 2017 - September 2020
B.Sc. in Statistics for Economics and Business Padua, Italy
Final Grade: 110/110, cum Laude (with honors)

Experience

RESEARCH VISIT
May 2025 - July 2025
University of Southern California Los Angeles, United States of America
Research at the CPS-VIDA (Cyber-Physical Systems: Verification, Intelligence, Design, and Analysis) Lab, under the supervision of Prof. Jyotirmoy V. Deshmukh. During this period, we focused on the automatic generation of challenging simulated testing scenarios for self-driving cars, by integrating deep generative models with spatio-temporal logic.

RESEARCH INTERNSHIP
April 2022 - June 2022
Vrije Universiteit Amsterdam, The Netherlands
Researching under the supervision of Prof. Zocca we developed a new model to generate synthetic power networks using random graph theory as well as both the statistical methods used to analyze the real world data to extract the information needed and a novel Monte-Carlo method to efficiently estimate samples from the proposed model

Theses topics

Bachelor thesis

Supervisor: Prof. Scarpa

Title: "Statistical Analysis of Open Data: the Case Study of AirBnB in Venice"
Main Topics: web scraping, statistics for geographically embedded data, methods for high dimensional statistics

Master thesis

Supervisors: Prof. Zocca, Prof. Formentin

Title: "Generating Synthetic Power Grids using Exponential Random Graph Models"
Main Topics: statistical model for network data, MCMC methods, random graph theory

Published papers

Model Abstraction and Conditional Sampling with Score-Based Diffusion Models — QEST 2023
L. Bortolussi, F. Cairoli, F. Giacomarra, D. Scassola

Generating synthetic power grids using exponential random graphs models — PRX Energy
F. Giacomarra, G. Bet, A. Zocca

Certified Guidance for Planning with Deep Generative Models — AAMAS25
F. Giacomarra, M. Hosseini, N. Paoletti, F. Cairoli

Demystifying Generative AI: A Pedagogical Approach. — Ital-IA 2025
E. Ballarin, F. Giacomarra, A. Mecchina, N.A. Pearson

Guiding Neuro-Symbolic Scenario Generation with Spatio-Temporal Logic — AAMAS26
L. Bonin, F. Giacomarra, F. Cairoli, J. Deshmukh

Attended conferences

AAMAS26

May 2026
Paphos, Cyprus

Oral presentation at the 25th International Conference on Autonomous Agents and Multiagent Systems

EURIPS25

December 2025
Copenhagen, Denmark

Poster presentation at the EURIPS conference

AAMAS25

May 2025
Detroit, USA

Oral presentation at the 24th International Conference on Autonomous Agents and Multiagent Systems

DTW 2025

February 2025
Padua, Italy

Oral presentation at the International Workshop on Digital Twins: Mathematical Analysis, Formal Methods and Scientific Machine Learning.

SMLET 2024

June 2024
Trieste, Italy

Poster presentation at the Scientific Machine Learning, Emerging Topics international conference.

Tutoring and mentoring

B.SC. THESIS SUPERVISOR

October 2025
Trieste, Italy

University of Trieste
Co-Supervisor for the B.Sc. thesis of Tonet Lorenzo. Title of the thesis: **Generation of synthetic graphs: comparison and synergies between probabilistic approaches and deep learning**

TUTOR

December 2024 - December 2025
Trieste, Italy

University of Trieste
Coordinator of the PLS tutors, responsible for activities aimed at reducing student dropout rates

TUTOR

December 2024 - February 2025
Trieste, Italy

University of Trieste
PLS Tutoring support to reduce student dropout rates

TUTOR

November 2020 - January 2021
Padua, Italy

University of Padua
Calculus tutor activity for students in B.Sc. in Statistics

Academic service

PC MEMBER	AAMAS27
REVIEWER	AAMAS, L4DC, IEEE TNNLS, IEEE T-ASE, Physical Review E
SUBREVIEWER	ECAI

Honors & awards

FAMELAB INTERNATIONAL AUDIENCE WINNER

2023

Cheltenham Festival
Biggest Science communication competition in the world. Selected during the international final as the audience winner by popular vote.

FAMELAB NATIONAL CHAMPION

2023

Fondazione PsiQuadro
Biggest national scientific communication contest. 1500€ final prize and access to the international final.

SCHOLARSHIP "MILLE E UNA LODE"

2018

University of Padua
1000€ scholarship awarded for the academic year 2017/2018 for being in the top 3% students

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